

positive association when restricted to witnessed, non-DOA out-of-hospital cardiac arrest cases, that subgroup is less applicable to the available baseline incidence rate of non-DOA out-of-hospital cardiac arrest cases.

Out-of-Hospital Cardiac Arrest

In a single-pollutant model of all non-DOA OHCA cases, the coefficient and standard error were estimated from a hazard ratio of 1.02 (95% CI: 0.92-1.12) for each 10 $\mu\text{g}/\text{m}^3$ increase in daily mean $\text{PM}_{2.5}$ concentrations, lagged by 0-1 days (Rosenthal et al. 2008, Table 5).

E.2.8 Silverman et al. (2010)

Silverman et al. (2010) investigated the link between short-term ambient air pollution exposure ($\text{PM}_{2.5}$, NO_2 , SO_2 , O_3 , and CO) and out-of-hospital cardiac arrest in New York City between 2002 and 2006. Authors obtained medical data from the Emergency Medical Services of the New York City Fire Department for 8,216 subjects aged 0 to 99, average age 65.6 with slightly more men than women. Authors collected air pollution and weather data from the US EPA's Air Quality System monitors within a 20-mile radius of New York City and averaged over 24-hour periods. Authors conducted time series and case crossover analyses with 0- and 1-day lagged air pollution levels and by season.

Out-of-Hospital Cardiac Arrest

In a single-pollutant case-crossover model, the coefficient and standard error were estimated from a relative risk of 1.04 (95% CI: 0.99-1.08) for a 10 $\mu\text{g}/\text{m}^3$ increase in the averaged daily mean $\text{PM}_{2.5}$ concentration 0- and 1-day prior to onset (Silverman et al. 2010, Table 4).

E.2.9 Sullivan et al. (2005)

Sullivan et al. (2005) studied the relationship between onset time of acute myocardial infarction and the preceding hourly $\text{PM}_{2.5}$ concentrations in 5,793 confirmed cases of myocardial infarction through King County, Washington. In this case-crossover study from 1988-1994, air pollution exposure levels averaged 1 hour, 2 hours, 4 hours, and 24 hours before onset of myocardial infarction were compared to a set of time-stratified referent exposures from the same day of the week in the month of the case event. The authors estimated that an associated risk of 1.01 (95% CI: 0.98-1.05) for myocardial infarction onset could be attributed to a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ the hour before the MI onset. No increased risk was found in all cases with preexisting cardiac diseases with an odds ratio of 1.05 (95% CI: 0.95-1.16). Furthermore, stratification for hypertension, diabetes, and smoking status did not modify the association between $\text{PM}_{2.5}$ and onset of myocardial infarction.

Acute Myocardial Infarction, Nonfatal

In a single-pollutant model the coefficient and standard error were estimated from the odds ratio (1.02) and 95% confidence interval (95% CI: 0.98-1.07) for a 10 $\mu\text{g}/\text{m}^3$ increase in daily 24-hour mean $\text{PM}_{2.5}$ lagged 1 day (Sullivan et al., 2005, Table 3).

Incidence Rate: AMI hospital admission rate for all ages. See the incidence rate discussion under Peters et al. (2001) in Section E.2.5.

Population: Ages 18 and older

Adjustment: See the adjustment description in Section E.2.5.

E.2.10 Wei et al. (2019)

Wei et al. (2019) evaluated the relationship between short-term $\text{PM}_{2.5}$ exposure and hospital admissions for 214 mutually exclusive disease groups, including acute myocardial infarction, in a time-stratified, case-crossover analysis of over 95 million Medicare inpatient hospital claims from 2000-2012. The authors estimated daily $\text{PM}_{2.5}$ levels at a 1-km² grid cell level using a satellite based, neural network model that was calibrated using monitor data and assigned 0-1 day lagged $\text{PM}_{2.5}$ exposure to each participant by zip code of residence. For each disease group, Wei et al. (2019) created a case crossover dataset that controlled for individual level and zip code level variables, day of the week, seasonality, and long-term time trends. They used conditional logistic regression models to estimate associations between $\text{PM}_{2.5}$ exposure and risk of hospital admission and found positive associations for numerous rarely studied and numerous well-studied disease groups.

Acute Myocardial Infarction, Nonfatal

In a single-pollutant model, the coefficient and standard error are estimated from a reported relative increase in risk (0.11%) and 95% confidence interval (0.07%-0.16%) associated with a 1 $\mu\text{g}/\text{m}^3$ increase in 0-1 day lagged $\text{PM}_{2.5}$ exposure (Wei et al. 2019, Figure 3, CCS 100 Acute Myocardial Infarction).

E.2.11 Zanobetti and Schwartz (2006)

Zanobetti and Schwartz (2006) analyzed hospital admissions through emergency department for myocardial infarction (ICD-9 code 410) and pneumonia (ICD-9 codes 480-487) for associations with fine particulate air pollution, ozone, black carbon, nitrogen dioxide, PM not from traffic, and CO in the greater Boston area from 1995-1999. The authors used a case-crossover analysis with control days matched on temperature. Significant associations were detected for NO_2 with a 12.7% increase (95% CI: 5.8-18.0), $\text{PM}_{2.5}$ with an 8.6% increase (95% CI: 1.2-15.4), and black carbon with an 8.3% increase (95% CI: 0.2-15.8) in emergency myocardial infarction hospitalizations. Similarly, significant associations were identified for $\text{PM}_{2.5}$ with a 6.5% increase (95% CI: 1.1-11.4) and CO with a 5.5% increase (95% CI: 1.1-9.5) in pneumonia hospitalizations.

Acute Myocardial Infarction, Nonfatal

The study looked at hospital admissions of AMI through the ER. Under the assumption that all heart attacks will end in hospitalization, we consider the endpoint as heart attack events to be consistent with other studies. In a single-pollutant model, the coefficient and standard error are estimated from the percent change in risk (8.65%) and 95% confidence interval (95% CI: 1.22-15.38%) for a 16.32 $\mu\text{g}/\text{m}^3$ increase in daily 24-hour mean $\text{PM}_{2.5}$ for an average of the 0- and 1-day lag (Zanobetti A. and Schwartz, 2006, Table 4).

Incidence Rate: AMI hospital admission rate for all ages. See the incidence rate discussion under Peters et al. (2001) in Section E.2.5.

Population: All ages. Note that although Zanobetti and Schwartz (2006) reports results for the 65-99 year old age range, for comparability to other studies, we apply the results to the population aged 18 and older. Since the vast majority of AMIs occur among population 65-99, over-counting may not be an issue when applying the risk coefficient to all ages.

Adjustment: See the adjustment description in Section E.2.5.

E.2.12 Zanobetti et al. (2009)

Zanobetti et al. (2009) examined the relationship between daily $\text{PM}_{2.5}$ levels and emergency hospital admissions for cardiovascular causes, myocardial infarction, congestive heart failure, respiratory disease, and diabetes among 26 U.S. communities from 2000-2003. The authors used meta-regression to examine how this association was modified by season- and community-specific $\text{PM}_{2.5}$ composition while controlling for seasonal temperature as a substitute for ventilation. Overall, the authors found that $\text{PM}_{2.5}$ mass higher in Ni, As, and Cr as well as Br and organic carbon significantly increased its effects on hospital admissions. For a 10 $\mu\text{g}/\text{m}^3$ increase in 2-day averaged $\text{PM}_{2.5}$, a 1.89% (95% CI: 1.34-2.45) increase in cardiovascular disease admissions, a 2.25% (95% CI: 1.10-3.42) increase in myocardial infarction admissions, a 1.85% (95% CI: 1.19-2.51) increase in congestive heart failure admissions, a 2.74% (95% CI: 1.30-4.20) increase in diabetes admissions, and a 2.07% (95% CI: 1.20-2.95) increase in respiratory admissions were observed. The relationship between $\text{PM}_{2.5}$ and cardiovascular admissions was significantly modified when the mass of $\text{PM}_{2.5}$ was high in Br, Cr, Ni, and sodium ions, while mass high in As, Cr, Mn, organic carbon, Ni and sodium ions modified the myocardial infarction relationship and mass high in As, organic carbon, and sulfate ions modified the diabetes admission rates.

Acute Myocardial Infarction, Nonfatal

The study looked at hospital admissions of AMI through ER. Under the assumption that all heart attacks will end in hospitalization, we consider the endpoint as heart attack events to be consistent with other studies. In a single-pollutant model the coefficient and standard error are estimated from the percent change in risk (2.25%) and 95%

confidence interval (95% CI: 1.10-3.42) for a 10 $\mu\text{g}/\text{m}^3$ increase in 2-day averaged $\text{PM}_{2.5}$ (Zanobetti et al., 2009, Table 3).

Incidence Rate: AMI hospital admission rate for all ages. See the incidence rate discussion under Peters et al. (2001) in Section E.2.5.

Population: All ages. Note that although Zanobetti et al. (2009) reports results for the 65-99 year old age range, for comparability to other studies, we apply the results to the population of aged 18 and older. Since the vast majority of AMIs occurs among population 65-99, over-counting may not be an issue when applying the risk coefficient to all ages.

Adjustment: See the adjustment description in Section E.2.5.

E.3 Hospitalizations

Table E-3 summarizes the health impacts functions used to estimate the relationship between $\text{PM}_{2.5}$ and hospital admissions. Below, we present a brief summary of each of the studies and any items that are unique to the study.

Table E-3. Core Health Impact Functions for Particulate Matter and Hospital Admissions

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Cardiovascular	Bell et al.	2015	213 U.S. Counties	65-99		D24HourMean	0.000648	0.000089	Log-linear	
Respiratory	Bell et al.	2015	213 U.S. Counties	65-99		D24HourMean	0.00025	0.000120	Log-linear	
Respiratory	Ostro et al.	2009	6 California counties	0-18		D24HourMean	0.002752	0.000772	Log-linear	

E.3.1 Bell et al. (2015)

Bell et al. (2015) investigated the effects of short-term fine particulate matter ($\text{PM}_{2.5}$) exposure on respiratory health (ICD-9 464-466, 480-487, 490-492, 493) and cardiovascular health (ICD-9 410, omitting 410.x2; 410-414; 426-427; 428; 429; 430-438; and 440-448) in older adults (>64 years). Authors acquired data for 213 U.S. counties (1999-2010) from the Medicare Claims Inpatient Files for U.S. residents >65 years of age. Authors chose variables including sex, age, county of residence, and cause of hospital admission, as determined by ICD-9 codes. Authors collected $\text{PM}_{2.5}$ exposure data from county population-based ambient monitors from the US EPA Air Quality System and averaged for county and day. Data were present for 56.5% of study days due to the sampling schedule of the monitors. Bell et al. (2015) utilized Bayesian hierarchical modeling to examine the links between $\text{PM}_{2.5}$ and hospital admissions, running separate models to generate risk models for time lags (0-2 days) and season for any estimated variation in health effects.